ModelIt!

Teacher's Guide

Can Soil Save the Climate?

Modeling Soil Carbon Sequestration, Regenerative Agriculture, and Feedback Loops

Quick Reference

Grade Level	Grade 11L2
NGSS Standard	HS-LS2-4, HS-ESS3-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS2-4, HS-ESS3-3

Use mathematical representations to support claims for the cycling of matter and flow of energy among organisms in an ecosystem. Create a computational simulation to illustrate the relationships among management of natural resources, the sustainability of human populations, and biodiversity.

Three-Dimensional Learning

Science Practice	**Using Mathematics and Computational Thinking** — Students use mathematical representations to model carbon flux rates between atmosphere, plants, and soil, calculating net sequestration under different management scenarios and evaluating whether soil carbon gains can meaningfully offset fossil fuel emissions.
Disciplinary Core Idea	**LS2.B Cycles of Matter and Energy Transfer / ESS3.C Human Impacts on Earth Systems** — Students model how matter cycles between living and nonliving components of the soil ecosystem, and how agricultural management decisions alter these cycles with consequences for atmospheric carbon concentration and climate stability.
Crosscutting Concept	**Stability and Change / Energy and Matter** — Students analyze how soil carbon exists in a dynamic equilibrium between inputs and outputs, and how human practices and climate change can shift this equilibrium, transforming soils from carbon sinks to carbon sources.

CAST Testing Alignment

MULTIPLE CHOICE:

Global soils contain approximately 2,500 gigatons of organic carbon. Agricultural soils have lost 50-70% of their original carbon through tillage. Regenerative agriculture can sequester 0.5-3 tons of carbon per hectare per year, but gains plateau after 20-30 years. Based on your model, which statement BEST describes soil carbon sequestration as a climate strategy?

CONSTRUCTED RESPONSE:

Using your model, explain the soil-climate feedback loop and why it represents one of the most dangerous tipping points in the climate system. Reference soil carbon stocks, microbial respiration rates, temperature sensitivity, and the concept of positive feedback in your explanation.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Beneath our feet lies the largest terrestrial carbon reservoir on Earth — soil holds more carbon than the atmosphere and all plant life combined. Understanding soil carbon dynamics requires modeling the intersection of plant biology, microbial ecology, agricultural science, and climate physics to evaluate whether changing how we manage soil could meaningfully slow climate change, or whether warming itself could turn this vast carbon store into an accelerant.

The Soil Carbon Reservoir

Global soils contain approximately 2,500 gigatons of organic carbon in the top 2 meters — more than three times the 880 gigatons in the atmosphere and four times the 450 gigatons in all living vegetation. This carbon accumulated over thousands of years as plant material was incorporated into soil and partially decomposed by microorganisms. The resulting soil organic matter (humus) is a complex mixture of compounds ranging from fresh plant residues to highly stable molecular aggregates that can persist for centuries. Soil organic carbon is not static — it exists in a dynamic equilibrium between inputs (plant growth, root exudation, leaf litter) and outputs (microbial respiration, erosion, leaching). The balance between these fluxes determines whether soil accumulates or loses carbon.

How Agriculture Destroyed Soil Carbon

When humans first plowed native grasslands and forests for agriculture, they initiated a massive transfer of carbon from soil to atmosphere. Tillage physically disrupts soil aggregates that protect organic matter from decomposition, exposes buried carbon to oxygen, and dramatically accelerates microbial breakdown. Globally, agricultural soils have lost an estimated 50-70% of their original organic carbon — approximately 133 gigatons of carbon — since cultivation began. The U.S. Midwest prairie soils, which originally contained 6-8% organic matter, now average 1-3%. This loss represents both an environmental catastrophe and an opportunity: the depleted carbon represents reservoir capacity that could theoretically be refilled.

Regenerative Agriculture and Carbon Rebuilding

Regenerative agriculture reverses the practices that deplete soil carbon. No-till farming eliminates the physical disruption that exposes organic matter to rapid decomposition. Cover crops — planted between cash crop seasons — maintain living roots in the soil year-round, continuously pumping carbon below ground through root growth and exudation. Diverse rotations support broader microbial communities that process carbon more efficiently. Compost and manure additions provide direct organic matter inputs. Research shows these practices can sequester 0.5-3.0 tons of carbon per hectare per year, depending on soil type, climate, and specific practices. However, gains are not permanent: they slow as soils approach a new equilibrium (typically 20-30 years), and they reverse rapidly if practices revert to conventional tillage.

The Dangerous Feedback Loop

The soil-climate feedback is among the most concerning positive feedback mechanisms in the climate system. Soil microbial respiration — the process by which microbes decompose organic matter and release CO2 — is highly temperature-sensitive. For every 10°C increase in temperature, respiration rates roughly double (the Q10 relationship). As global temperatures rise, soil microbes decompose organic matter faster, releasing more CO2, which causes more warming, which accelerates more decomposition. Arctic permafrost — containing approximately 1,500 gigatons of frozen carbon — is particularly vulnerable. As permafrost thaws, previously frozen organic matter becomes available to microbes for the first time in thousands of years. Models suggest that permafrost carbon release alone could add 0.3-1.0°C of additional warming by 2100 — a feedback that is essentially irreversible on human timescales.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L

Locate the System

Students identify six components of the soil carbon system: Agricultural Management Practice, Plant Carbon Fixation Rate, Soil Microbial Activity, Soil Organic Carbon Stock, Soil Respiration Rate, and Net Soil Carbon Balance — distinguishing the human-controlled management decisions from the biological and chemical processes that govern carbon cycling.

E

Establish Relationships

Students determine that Agricultural Management Practice controls Plant Carbon Fixation Rate through crop selection and soil disturbance levels. Plant inputs feed Soil Microbial Activity, which processes carbon into stable organic matter (increasing Soil Organic Carbon Stock) or respires it back to CO2 (Soil Respiration Rate). The balance between these processes determines Net Soil Carbon Balance.

V

Visualize & Model

Students build a computational model showing the stock-and-flow dynamics of soil carbon — the slow accumulation through plant inputs and microbial stabilization versus the rapid loss through tillage disturbance and temperature-driven respiration.

E

Evaluate Outcomes

Students run conventional baseline, regenerative transition, and climate warming feedback scenarios to quantify sequestration potential, identify saturation timescales, and test whether warming can flip soils from sinks to sources.

R

Revise & Extend

Students add Permafrost Carbon Pool, Soil Biodiversity Index, or Water Holding Capacity to explore how Arctic thaw amplifies global feedback, how biological diversity affects carbon processing efficiency, and how carbon sequestration creates water management co-benefits.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"There's more carbon under your feet than in the entire sky above you. And how we treat this ground determines whether that carbon stays put or joins the atmosphere."*

Watch For

Hold up two jars: one with pale, compacted conventional farm soil and one with dark, crumbly compost-rich soil. Pass them around. Ask: Which one do you think holds more carbon? Which one grew more food? The dark soil does both.

Slide 2: Learning Objectives

3 min

Say: *"Today you're modeling the biggest carbon reservoir most people have never thought about — and whether farming differently could help save the climate."*

Watch For

Have students read objectives. Pre-teach soil carbon by weight — a healthy soil is 5% organic matter by weight, half of which is carbon. A single hectare of healthy topsoil can store 100+ tons of carbon. Quick-write: If soil can store carbon, can it also release it? When?

Slide 3: Big Question

3 min

Say: *"Soils hold 2,500 gigatons of carbon. The atmosphere holds 880 gigatons. If we could move just 10% more carbon into soil, that's 250 gigatons — roughly 7 years of total global emissions. Could farming be the answer?"*

Watch For

Write the numbers on the board: Soil = 2,500 Gt. Atmosphere = 880 Gt. Annual emissions = 36 Gt. Have students calculate: what percentage of annual emissions would need to be sequestered in soil to offset them? Is that realistic? Let the tension build between hope and math.

Slide 4: LEVER Framework

2 min

Say: *"We're modeling a system where the input is slow and steady — plants growing — and the output can be fast and catastrophic — warming soils releasing their carbon. The question is which force wins."*

Watch For

Preview each LEVER step. Introduce the bathtub metaphor: soil carbon is the water level, plant inputs are the faucet (slow), respiration is the drain (temperature-controlled). What happens when someone turns up the heat on the drain?

Slide 5: Activity 1: Components

8 min

Say: *"One human choice — how we farm. Five system responses. But here's the twist: temperature controls the biggest output, and we're warming the planet."*

Watch For

Guide sorting of external vs. internal components. Only Agricultural Management Practice is external. Discuss why Soil Respiration Rate is internal — it responds to temperature and moisture conditions that humans influence indirectly through climate change. This indirect control makes the feedback loop so dangerous.

Slide 6: Activity 2: Connections

8 min

Say: *"Follow the carbon: atmosphere to plant, plant to soil, soil back to atmosphere. The balance between in and out determines everything. And temperature is tipping that balance."*

Watch For

Help students map the carbon flux pathways. Identify the critical feedback loop: warming increases respiration, respiration releases CO2, CO2 causes more warming. Ask: Is this a positive or negative feedback? What makes it so dangerous? (Positive — self-reinforcing, no natural brake.)

Slide 7: Activity 3: Simulation

10 min

Say: *"Three scenarios: keep depleting the soil, rebuild it with regenerative farming, or rebuild it while the planet warms. The third scenario is the real test — can soil gains outrun temperature losses?"*

Watch For

Students predict before running simulations. Key focus: the saturation curve in the regenerative scenario (rapid gains at first, then plateau) and the crossover point in the warming scenario where respiration exceeds fixation. Compare Net Soil Carbon Balance across all three.

Slide 8: Discoveries

5 min

Say: *"Your model just showed that soil is both a potential solution and a potential time bomb. Regenerative agriculture can help — but it cannot outrun unchecked warming. The carbon already in the soil is the real stakes."*

Watch For

Lead discussion on realistic versus overhyped claims about soil carbon. Some advocates claim soil can 'solve' climate change; the model shows it can contribute but has limits. Discuss the permafrost feedback — 1,500 Gt of frozen carbon beginning to thaw. What happens if that enters the atmosphere?

Slide 9: STEM Challenge

12 min

Say: *"A state wants to convert 100,000 hectares to regenerative farming and count the carbon. Design the program that makes it happen — the practices, the measurement, and the money."*

Watch For

Groups design comprehensive transition programs including practice recommendations by soil type, carbon measurement and verification protocols, farmer incentive structures (carbon credits, subsidies, technical assistance), and 25-year projections. Present as proposals to a state agricultural board.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Agricultural Management Practice
- **INTERNAL:** Plant Carbon Fixation Rate, Soil Microbial Activity, Soil Organic Carbon Stock, Soil Respiration Rate, Net Soil Carbon Balance

Relationships

- **Agricultural Management Practice to Plant Carbon Fixation Rate:** POSITIVE (+) for regenerative practices — Regenerative practices like cover cropping and diverse rotations increase the total amount of plant photosynthesis occurring on farmland by maintaining living roots year-round rather than leaving soil bare between cash crops. No-till also preserves root systems that pump carbon underground.
- **Plant Carbon Fixation Rate to Soil Microbial Activity:** POSITIVE (+) — Higher plant carbon inputs — through root exudates, dead root tissue, and leaf litter — provide more food for soil microorganisms. Mycorrhizal fungi receive up to 30% of plant-fixed carbon directly through symbiotic root connections, feeding the soil food web.
- **Soil Microbial Activity to Soil Organic Carbon Stock:** COMPLEX (+ and -) — Microbial activity has a dual role: microbes stabilize some carbon into long-lasting organic compounds (humus formation, mineral association) while simultaneously respiring other carbon back to CO2. The net effect depends on microbial community composition, soil mineralogy, and environmental conditions.
- **Temperature to Soil Respiration Rate:** POSITIVE (+) — Soil microbial respiration approximately doubles for every 10°C increase in temperature (Q10 ≈ 2). As global temperatures rise, microbes decompose soil organic matter faster, releasing more CO2. This is the mechanism that drives the dangerous soil-climate positive feedback loop.
- **Soil Respiration Rate to Net Soil Carbon Balance:** NEGATIVE (-) — Higher respiration removes carbon from the soil and returns it to the atmosphere as CO2. When Soil Respiration Rate exceeds Plant Carbon Fixation Rate, Net Soil Carbon Balance turns negative — the soil becomes a net carbon source rather than a sink.

Key Discoveries

- The world's soils contain approximately 2,500 gigatons of organic carbon — more than three times the carbon in the atmosphere (880 Gt) and four times the carbon in all living vegetation (450 Gt) combined
- Agricultural soils have lost 50-70% of their original organic carbon through centuries of tillage, which exposes buried carbon to oxygen and accelerates microbial decomposition — this represents a massive potential reservoir for re-sequestration
- Regenerative agriculture can sequester 0.5-3.0 tons of carbon per hectare per year, but gains slow as soils approach a new equilibrium — most sequestration occurs in the first 20-30 years after practice change, then plateaus
- The soil carbon-climate feedback is one of the most dangerous positive feedback loops in the climate system: warming accelerates microbial respiration, releasing stored carbon, which causes more warming, which accelerates more respiration — potentially releasing hundreds of gigatons of carbon from thawing permafrost alone

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Soil can absorb unlimited carbon and solve climate change"	Soil carbon sequestration has a physical saturation point determined by soil mineralogy, clay content, and climate. Once soil reaches its carbon carrying capacity — typically after 20-30 years of regenerative management — it can no longer accumulate additional carbon regardless of management intensity. Furthermore, even at maximum global sequestration rates, soil can offset only 5-10% of annual fossil fuel emissions. Soil carbon is a valuable tool in the climate toolkit, not a silver bullet.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"No-till farming automatically increases soil carbon"	No-till reduces carbon loss from tillage disturbance but does not guarantee net carbon gain. Without additional practices — cover crops, diverse rotations, organic matter additions — no-till may simply redistribute carbon within the soil profile rather than increasing total stocks. Some studies show no-till increases carbon in the top 0-10 cm but decreases it at 10-30 cm depth. The full regenerative package is needed for reliable sequestration.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Soil carbon, once stored, stays stored permanently"	Soil carbon is inherently impermanent — it exists in a dynamic equilibrium that can shift rapidly. A single pass with a moldboard plow can expose decades of accumulated organic matter to rapid decomposition, releasing stored carbon within a few years. Climate warming independently accelerates decomposition rates. Drought can kill plants and reduce inputs. This permanence problem is the central challenge of soil carbon as a climate strategy — carbon credits based on soil storage must account for the risk that the carbon will be re-released.